

Hands-on Lab: Build a Library Management REST API

Objective: Build a simple Library Management REST API using Spring Boot and an in-memory list (no database).

Required Tasks

- Create a Book model (id, title, author, category).
- Create a BookService to manage books in memory.
- Create a BookController with endpoints: GET /books, GET /books/{id}, POST /books, DELETE /books/{id}.
- Configure the application using application.yml.
- Create configuration using @ConfigurationProperties.
- Define a custom bean using @Bean.
- Add logging using SLF4J.
- Create dev and prod profiles.
- Write unit tests for the service layer.

Bonus Challenge

- Implement PUT /books/{id}.
- Write Mockito unit tests for BookController.
- Write Web Layer tests using @WebMvcTest and MockMvc.

Concepts Covered

- Project Structure
- IoC & Dependency Injection
- Spring Beans
- REST Controllers
- Configuration
- Profiles
- Logging
- JUnit
- Mockito
- Web Layer Testing